Adding a glucose-indexed sensitivity curve to open-source automated insulin delivery

Initial proposal, with the TDD-based dynamic ISF in AndroidAPS and Trio as the comparator

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Status

This is a draft for discussion. Nothing described here has run on a device or in a simulator, and nothing was fitted.

The model proposed is the curve of sensitivity against glucose published by Romero-Ugalde and colleagues in 2025 [1], linked to total daily dose (TDD). The shape is theirs. The link to TDD is derived in this document and appears nowhere in the publication.

An earlier draft is in git (commit fd3662e). It proposed a sensitivity anchored to the entered profile and flat above 100 mg/dL, and is kept in the code as one arm of the evaluation.

Glucose is in mg/dL throughout because the code is. 100 mg/dL is 5.6 mmol/L and 200 mg/dL is 11.1 mmol/L.

Summary

The curve model and the existing dynamic ISF have the same form. Each is a function of glucose divided by a daily dose:

	Sensitivity	**Value of the glucose function at 100, 150, 200, 300 mg/dL**
Existing	1800 / ln(glucose / divisor + 1), over TDD	1737, 1368, 1173, 965 (divisor 55)
Curve model		4378, 2492, 1674, 1148

	Sensitivity	Value of the glucose function at 100, 150, 200, 300 mg/dL
	1915 × Q(glucose) / Q(180), over TDD	

Q is the published curve in mg/dL per unit. 1915 is the product of entered sensitivity and daily dose measured in adults in this series (document 6), taken to hold at 180 mg/dL, the reference glucose the publication uses. That choice sets the level and is a modelling choice.

Replacing one equation with the other is a small change to the code: the glucose function is swapped and the insulin-type divisor goes. The change in dosing is large. At the same TDD the curve asks for less insulin than stock AndroidAPS at every glucose up to 362 mg/dL with a divisor of 55, by a factor of 2.52 at 100 mg/dL, 1.43 at 200 and 1.19 at 300. It falls by 2.61 from 100 to 200 mg/dL where the existing equation falls by 1.48.

A second way of scaling the curve is carried as an alternative: the entered profile value in place of 1915 / TDD. It suits Trio, which already expresses dynamic ISF as a ratio of the profile, and it does not move with recent dosing.

Most of what has to be added sits around the equation. The level is the least settled part, because the publication gives no daily dose: three ways of setting it give reference doses of 24, 54 and 71 to 78 U. A decision is needed on where along the curve a correction is read. Trio's default limits would remove most of the curve.

The evidence is set out in its own section. The curve is reproducible. The analysis in this series does not support reading its decline between 150 and 250 mg/dL as a change in sensitivity, and whether dosing to it does better than the existing equation has not been tested. The path proposed is therefore replay and simulation first, then logging beside the existing equation, with dosing only after those.

The model

The published curve

The publication [1] used records from 11,201 adults with type 1 diabetes on the DBLG1 closed-loop system. One event, as it defines them: glucose peaks at 240 mg/dL at least an hour after a meal, with under 2.5 g of carbohydrate still on board, and falls to a minimum of 160 mg/dL more than 20 minutes later. Between the two, insulin on board dropped and some insulin was

delivered above the reference basal, 2.0 U in all. The sensitivity for that event is 80 / 2.0 = 40 mg/dL per unit. The window ends where glucose stopped falling, so events are selected on their outcome and the figure is a ratio of outcomes.

The reported result is a sensitivity that rises from 65 to 79 mg/dL per unit between 50 and 100 mg/dL, falls to 35 at 180 and is below 20 above 300. Against a dose sized at 180 mg/dL, the authors put the dose at about half for glucose up to 120 mg/dL and about double above 340.

For glucose above 100 mg/dL the curve is carried here as a quartic, recorded in this series in March 2026 from the conference poster (`inv008.dynisf.quartic_isf`). It gives 80.6 at 100 mg/dL, 35.3 at 180 and 21.1 at 300, and dose multiples of 0.56 at 120 and 1.95 at 340, in line with the abstract. The poster's separate curve for glucose below 100 mg/dL was not recorded.

Linking it to daily dose

The published values are absolute. A person on 20 U a day and a person on 80 U a day would get the same 35 mg/dL per unit at 180 mg/dL, so the curve needs a scale for each person before it can dose. Two links are set out.

The TDD link, which is the candidate:

sensitivity = 1915 × Q(glucose) ÷ Q(180) ÷ TDD

The profile link, carried as an alternative:

sensitivity = profile sensitivity × Q(glucose) ÷ Q(180)

For a person on 40 U a day with a profile of 50 mg/dL per unit, the TDD link gives 109 mg/dL per unit at 100 mg/dL, 62 at 150, 48 at 180, 42 at 200 and 29 at 300. The profile link gives 114, 65, 50, 44 and 30. The two agree for anyone whose profile times daily dose is 1915, and differ by that ratio for everyone else.

	TDD link	**Profile link**
Fits	AndroidAPS, where the profile value is unused in dynamic mode	Trio, where dynamic ISF is a ratio of the profile
Entered sensitivity and its schedule through the day	Unused	Kept
Recent dosing	Moves the value, as now	No effect
What has to be chosen		

	TDD link	Profile link
	The level (1915) and the definition of TDD	The glucose at which the profile is taken to hold

The reference implementation (`inv013/candidate.py`) anchors both links at 180 mg/dL, a modelling choice made because that is the publication's own reference point. Its treatment of the ends is a modelling choice as well. Below 100 mg/dL the curve is held at its value at 100, which gives less insulin there than the published curve, since the published one falls towards 65. Above 400 mg/dL it is held at its value at 400, where the published range ends.

The level

The publication reports no daily dose, so the level cannot be taken from it. The reference dose is the daily dose at which the published values would apply unchanged, and it follows from whichever level is assumed:

Level assumed	Reference dose, U	Curve × TDD at 200 mg/dL
Adult entered product, 1915, at 180 mg/dL (this draft)	54.3	1,674
The 1800 rule at 105 mg/dL (this series, March 2026)	23.7	731
The same construction on Loop, Trio and oref0 records, at 200 mg/dL	70.8 to 77.8	2,182 to 2,398

These differ by a factor of three. The first asks that a person's entered sensitivity be right at 180 mg/dL. The second asks that it be right near target and puts the whole curve at less than half the level. The third is the level measured by the method on open-source records, 30 to 43% above this draft's, and it comes from event rules that differ from the published ones. The existing adjustment factor moves the level for one person in either direction. The default is the open point.

The comparator

The comparator is the dynamic ISF in stock AndroidAPS (upstream master f77abd9b82, 31 October 2026) and stock Trio (upstream main 0b29cd534, 7 October 2026), read from source on 10 November 2026. oref0 as published has autosens and no TDD-based dynamic ISF, so it has nothing to replace. Loop and iAPS were not read.

AndroidAPS (`OpenAPSSMBPlugin.calculateRawDynIsf`, `DetermineBasalSMB.kt`):

- The dose term is 0.33 of a weighted eight hour total scaled to a day, 0.34 of the seven day average and 0.33 of the most recent day, multiplied by the adjustment factor (default 100%, range 1 to 300%) and the profile percentage.
- The divisor is 55, 65 or 75 according to the insulin peak time, which moves the level at 100 mg/dL by 22% between the first and last.
- In dynamic mode the dose calculation uses this value in place of the profile sensitivity, so the entered value and its schedule through the day are unused.
- Each predicted glucose in the three prediction arrays gets its own sensitivity from the same equation. The dose is sized on a further value calculated at a glucose chosen by one of three branches.
- With any of the five dose totals missing, the plugin falls back to the profile and raises a notification.

Trio (`DynamicISF.swift`, a Swift port of the `trio-oref` JavaScript):

- Logarithmic mode expresses the same equation as a ratio of the profile sensitivity, with a default adjustment factor of 0.8, and holds the ratio inside the autosens limits (defaults 0.7 and 1.2). The result is bounded between profile / 1.2 and profile / 0.7.
- Sigmoid mode is 1 at the profile target, so its level is the profile value, and it is bounded by the same limits.
- The dose term is 0.65 of the last two hours' average and 0.35 of the ten day average by default.

Side by side

One person: profile sensitivity 50 mg/dL per unit, 40 U a day, rapid-acting insulin, recent dosing equal to their average. Values in mg/dL per unit, from `results/inv013_comparator.md`.

Glucose, mg/dL	AndroidAPS, divisor 55	Trio logarithmic	Trio sigmoid	Curve, TDD link	Curve, profile link	Profile
80	50.1	62.6	53.7	109.4	114.3	50.0
100	43.4	54.3	50.0	109.4	114.3	50.0
120	38.9	48.6	47.1	86.2	90.1	50.0
150	34.2	42.8	44.3	62.3	65.1	50.0

Glucose, mg/dL	AndroidAPS, divisor 55	Trio logarithmic	Trio sigmoid	Curve, TDD link	Curve, profile link	Profile
180	31.0	41.7	42.9	47.9	50.0	50.0
200	29.3	41.7	42.4	41.9	43.7	50.0
250	26.3	41.7	41.8	33.7	35.1	50.0
300	24.1	41.7	41.7	28.7	30.0	50.0
400	21.3	41.7	41.7	19.9	20.8	50.0

Against each comparator the TDD link moves dosing differently:

- Against stock AndroidAPS it gives less insulin up to 362 mg/dL with a divisor of 55, 339 with 65 and 313 with 75, by most near target. Above those it gives more.
- Against Trio at its defaults it gives less insulin below about 200 mg/dL and more above, because Trio's upper limit holds its value at 41.7 by 180 mg/dL.
- Against this person's profile it gives less insulin below 174 mg/dL and more above.

Reading the curve for a dose

The abstract does not say which glucose an event is filed under. This draft takes it to be the glucose halfway between the peak and the minimum, which is an assumption. The existing code reads its equation at the current glucose, or at a glucose chosen by one of the three branches. For a flat sensitivity the two are the same. For a curve that falls this steeply they differ.

Units to bring the same person to 100 mg/dL, by the sensitivity alone (insulin on board, predictions and delivery limits are left out):

Starting glucose, mg/dL	AndroidAPS, divisor 55	Trio logarithmic	Profile	Curve, read at start	Curve, read at midpoint
150	1.46	1.17	1.00	0.80	0.61
200	3.41	2.40	2.00	2.39	1.61
250	5.71	3.60	3.00	4.46	3.01
300	8.29	4.80	4.00	6.97	4.78

Reading the curve at the starting glucose asks for 31 to 49% more insulin than reading it at the midpoint. From 200 mg/dL the difference is 2.39 against 1.61 U, on either side of the profile's 2.00. Integrating along the path gives 1.62 U, close to the midpoint reading.

This is a decision the curve model forces and the existing equation did not. The proposal's default is the midpoint between the glucose being corrected and the target, a modelling choice. The reading at the starting glucose is carried as an arm.

What has to be added

Component	Existing	Curve model	To add
Glucose function	1800 / ln(glucose / divisor + 1)	The published quartic over its value at 180 mg/dL	One function per system with shared test vectors. The published curve below 100 mg/dL, from its authors
Insulin-type divisor	55, 65 or 75	None	Removed in this mode
Level	1800	1915 at 180 mg/dL	A default, chosen after stage 2 below. The three candidates span a factor of three
Daily dose	AndroidAPS blend of five totals. Trio weighted two hour and ten day averages	Not defined by the publication	Default: the existing blend, so that this half of the behaviour is unchanged. Arm: the plain average
Level for one person	Adjustment factor	The same	The factor stays. Its guidance has to be worked out again, since at 100% the value at 100 mg/dL is 2.5 times the existing one
Profile link	AndroidAPS none. Trio a ratio of the profile	Profile × Q(glucose) / Q(180)	A second mode, or the Trio form of the first. A decision for the maintainers
Where the curve is read	Current glucose and three branches	Not stated in the publication	A reading rule, tested in replay
Below 100 and above 400 mg/dL	The equation continues	Published curve falls below 100. No data above 400	Held at the end values. Arm: the existing logarithm's rise below 100 mg/dL
Limits	AndroidAPS none. Trio 0.7 to 1.2 around the profile	Runs from 0.40 to 2.19 of the worked person's profile	A decision for Trio. Inside its default limits the curve runs only from 137 to 201

Component	Existing	Curve model	To add
			mg/dL for the worked person
Predictions	Sensitivity at each predicted glucose	Rises 2.61 times as predicted glucose falls from 200 to 100 mg/dL	The same call sites. Replay of the effect on minimum predicted glucose and on time at zero basal
Stored value	`variableSens` in the loop result		Both values stored while the existing one doses
Settings	Adjustment factor and mode options		A mode selector. No new number shown to the person in the first version
Tests	`ReplayApsResultsTest` with JavaScript parity assets. `DynamicISFTests`		Unit tests on the reference vectors, replay assets for the new mode, a guard test that the existing mode's output is unchanged

Code change points. AndroidAPS: `calculateRawDynIsf`, the three per-prediction lines and the three dose-sensitivity branches in `DetermineBasalSMB.kt`, and the preference screen. Trio: `DynamicISF.swift`, the forecast adjustment used in logarithmic mode, and the `trio-oref` JavaScript it was ported from.

What is known about the curve

The reproduction in this series (INV-012) was built before the abstract was read and its event rules differ: three hours clear of a meal where the publication uses one hour with under 2.5 g on board, and a peak that stands 15 mg/dL clear where the publication uses a 20 minute interval. It also multiplied each event by the person's TDD. The rows below compare ratios, which do not depend on that.

Question	Finding	Source
Is the curve reproducible?	A similar construction on 1,790 people in eight cohorts gave a ratio of 2.2 to 4.0 between 100 and 200 mg/dL, against 2.61 in the published curve. With no algorithm	Document 5 section 7, INV-012

Question	Finding	Source
	dosing (REPLACE-BG) it was 2.55 [2.25, 2.82]	
Does the level carry across systems?	At a midpoint of 200 mg/dL the construction gave 2,182 under Loop, 2,273 under Trio, 2,398 under oref0 and 3,283 in REPLACE-BG, against 1,674 from this draft's link. TDD definitions were not matched between sources	`results/inv012_peak_trough.md`
What does the method return for a constant sensitivity?	Simulated people with a true value of 2,000 at every glucose returned a falling curve reading 1.4 to 3.4 times the true value. Their ratio from 200 to 250 mg/dL was 0.87 [0.83, 0.90] and 0.92 [0.89, 0.95], against 0.84 to 1.05 in the six trial cohorts and 0.80 in the published curve	Document 5 section 7
What produces the decline between 100 and 200 mg/dL?	The median fall is 30 to 43 mg/dL at 100 and 39 to 64 at 200. The median insulin used is 0.02 to 0.2 U at 100 and 0.3 to 1.5 U at 200. At 100 mg/dL, 9 to 58% of events had no net insulin used. A fitted fall of 27 to 44 mg/dL at zero net insulin reflects basal and earlier doses acting, and the choice of events that end at a minimum	Document 5 section 7, `results/inv012_peak_trough.md`
Above 200 mg/dL?	Fall per extra unit at 300 against 200 mg/dL was 0.66 [0.49, 0.73] in REPLACE-BG (24 people) and 0.73 [0.64, 0.93] under Loop (49 people), against 0.69 in the published curve. No simulated arm reached 300 mg/dL	Document 5 section 7
Has the shape been tried as a predictor on oref records?	On 80 people and 62,344 overnight windows, with each shape anchored to the profile at 100 mg/dL, the typical miss was 24.2 mg/dL for the static profile and 29.8 for this shape. At 205 to 260 mg/dL it was 37.8 for the static profile and 73.7 for this shape	`oref-investigations/results/bridge_diabeloop.md`

Four limits apply to that table. The simulation did not reach 100 or 300 mg/dL, so the steep part of the curve near target has not been reproduced from a constant sensitivity and the case there rests on the split between fall and insulin used. The overnight windows in the last row were selected on the later outcome, as the audit of 16 September 2026 recorded, and that test anchored the shape at 100 mg/dL where this draft anchors at 180. Every row is observational. And the publication is a conference abstract from the system's manufacturer, with no full paper found.

What follows for the proposal is narrower than a verdict on the curve. As a description of net fall per net unit in events chosen from peak to minimum, it is reproducible on other people's data. As a statement that a unit does less at higher glucose, it is not supported between 150 and 250 mg/dL and is open above that. Whether a loop that doses to it does better than one dosing to the existing equation or to a flat profile is a separate question, and no evidence in this series bears on it directly.

A June 2026 proposal that used this curve's shape (document 3) was withdrawn for live dosing on 16 September 2026. The audit left such candidates open to simulation, shadow replay and prospective study. The path below stays inside that.

Evaluation path

Arms: the existing equation as shipped; the profile alone; the TDD link read at the midpoint; the TDD link read at the starting glucose; the TDD link on plain average TDD; the profile link; and the profile-anchored form of this document's first draft, which is flat above 100 mg/dL and is the arm the curve has to beat for its shape to matter.

Stage	What is done	What it decides
1. Replay of the equation	Each arm's value calculated on recorded histories (the INV-008 pipeline), split by glucose, by time since the last meal and by whether meals were announced	How much insulin each arm asks for, in whom and when. Descriptive, no pass rule
2. Curve on each system	`inv012/peak_trough.py` run on AndroidAPS and Trio records under the published event rules, with the simulated constant-sensitivity arms extended to 100 and 300 mg/dL	The default level, and whether the decline near target and above 250 mg/dL appears without a real effect

Stage	What is done	What it decides
3. Simulation	Paired runs by seed in TimSim, with arms for unannounced meals and for a mis-set adjustment factor	A two-sided harm rule written before the runs: time below 54 mg/dL, and time above 250 mg/dL, since against stock AndroidAPS this model gives less insulin over most of the range
4. Shadow running	Both values calculated on the device each cycle, the existing one doses, both are stored	The distribution of the ratio per person and the starting factor
5. Use	The developer's own device first, then released lines	Follows only after stages 3 and 4 pass. Stops on the harm rule

Stages 1 and 2 are analysis and carry the analysis checklist. Stages 4 and 5 change code that can change a dose, so they carry unit and guard tests, the repository's review checklist, a replay against real history and a fresh-context review of the diff.

The way back at every stage is the mode selector. The existing equation stays in the code unchanged and remains the default until stage 5 has run, and a build that includes the new mode names the previous build.

Definitions to fix

- The level, and with it the reference dose.
- Which daily dose the curve is divided by, and over how many days.
- How the reference basal is defined, and which insulin action model gives the insulin on board. The abstract states neither and the level depends on both.
- Which glucose an event is filed under. This draft assumes the midpoint.
- The curve below 100 mg/dL, which the publication describes and this repository does not hold.

Open questions

- Whether the published curve may be built into open-source code. It was derived by a manufacturer from its own users' records, and its authors have not been asked.
- Whether a level set from entered settings at 180 mg/dL applies to any open-source system. The construction on those systems' records reads 30 to 43% higher at 200 mg/dL.

- Whether the steep part between 100 and 150 mg/dL should be used for dosing at all, given that a large share of events there had no net insulin used.
- How the bolus calculator in each system takes its sensitivity in dynamic mode, and what stock autosens does there. Neither was read.
- Whether TimSim's oref adapter runs the existing dynamic ISF closely enough to be the comparator in stage 3. This was not checked.
- Whether the maintainers of AndroidAPS and Trio want one shared definition. Their two implementations of the existing equation already differ by 42% for the worked person at 200 mg/dL.

Reproducing the numbers

`python3 -m inv013.candidate` writes `results/inv013_comparator.md`, which holds every number in the comparison tables and the reference doses. `inv013/tests/test_candidate.py` checks the stock equation against the replay code of INV-008, the curve against the values stated in the abstract, and the ways of reading it. All other figures are quoted from the documents named beside them. The worked event in "The published curve" is an illustration with round numbers and is no one's record.

Reference

1. Romero-Ugalde HM, Adenis A, Cartier-Michaud T, Gauthier P, Desir C, Le Roux-Mallouf T, Huneker E, Benhamou PY. 920-P: The Relationship between Glycemia and Insulin Sensitivity Factor. Diabetes 2025;74(Supplement 1):920-P. doi:10.2337/db25-920-P